

4. Multimodal Visual Understanding + Visual Line Following

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Introduction

In this tutorial, we will learn how to issue commands via chat with Muto in the Dify interface.

Operation Steps

Preparatory Steps

This lesson requires that the Dify terminal be launched beforehand, and all configuration items be configured as in the previous section.

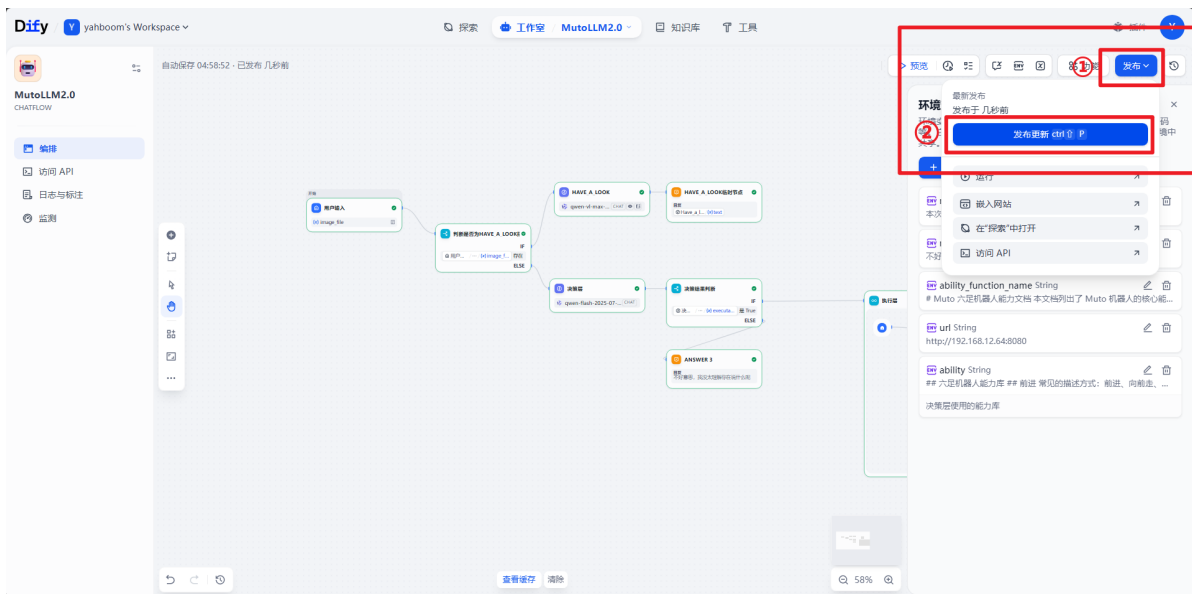
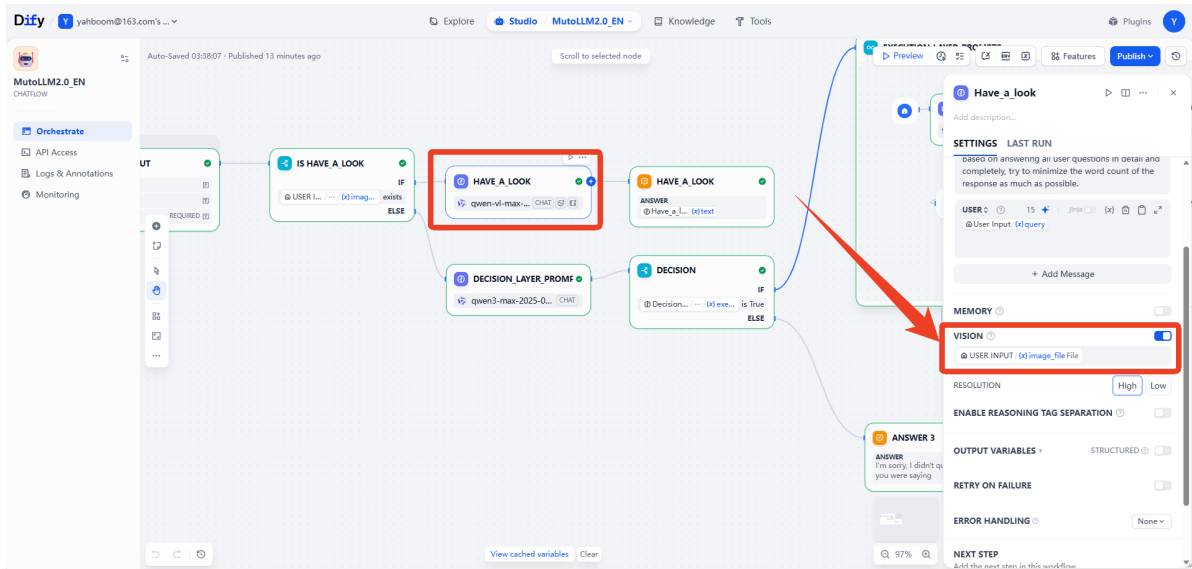
The image shows a terminal window with a bash shell and the Dify interface. In the terminal, the following commands are shown:

```
1 #!/usr/bin/env bash
2
3 # =====
4 # User Configuration Area
5 # =====
6
7 # 1. Voice Module Configuration
8 # =====
9
10 # 1.1 Provider & Keys
11 export ALIBABA_API_KEY="sk-XXXXXXXXXXXXXXXXXXXX20"
12
13 export FORCE_VOICE_PROVIDER="xunfei"
14
15 # 1.2 General Voice Settings
16 export VOICE_LANGUAGE="en"
17
18 # 2. Dify Configuration
19 # Auto-detect IP from wlan0
20 export DIFY_HOST=$(ip addr show wlan0 2>/dev/null | grep "inet " | awk '{print $2}' | cut -d/ -f1 | head -n 1)
21
22 # Fallback if wlan0 has no IP
23 if [ -z "$DIFY_HOST" ]; then
24   export DIFY_HOST="127.0.0.1"
25 fi
```

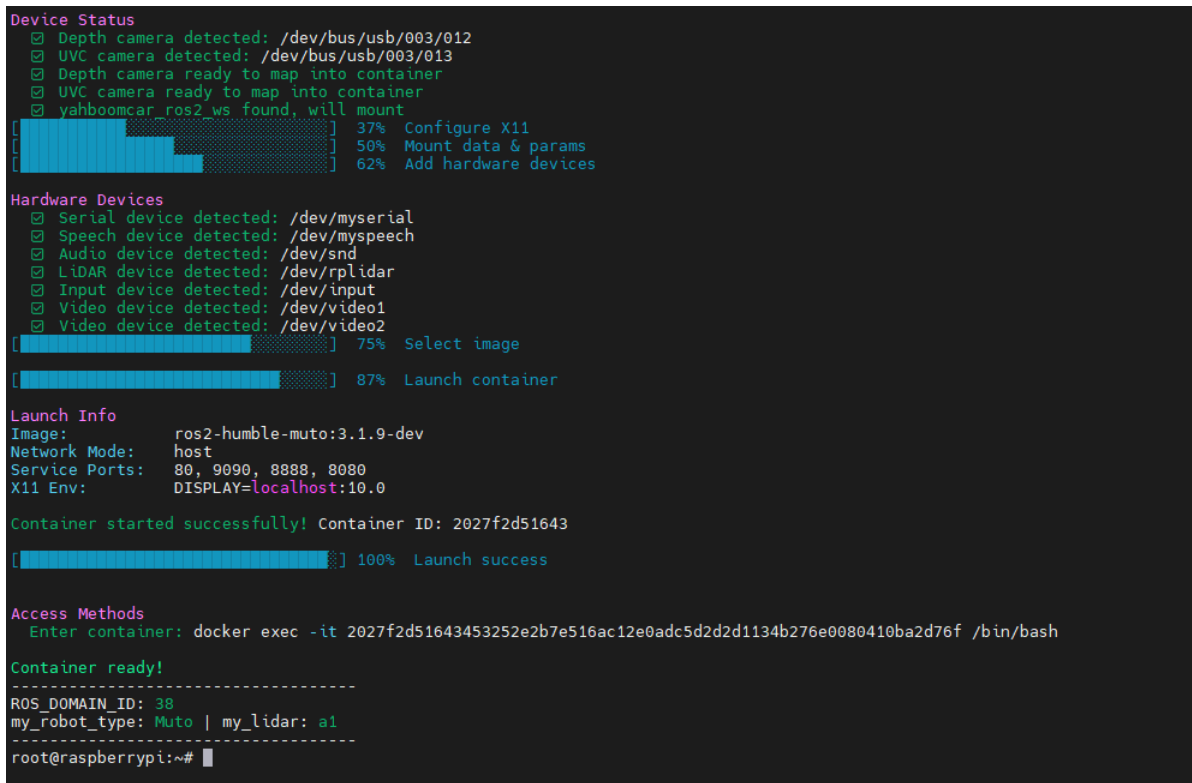
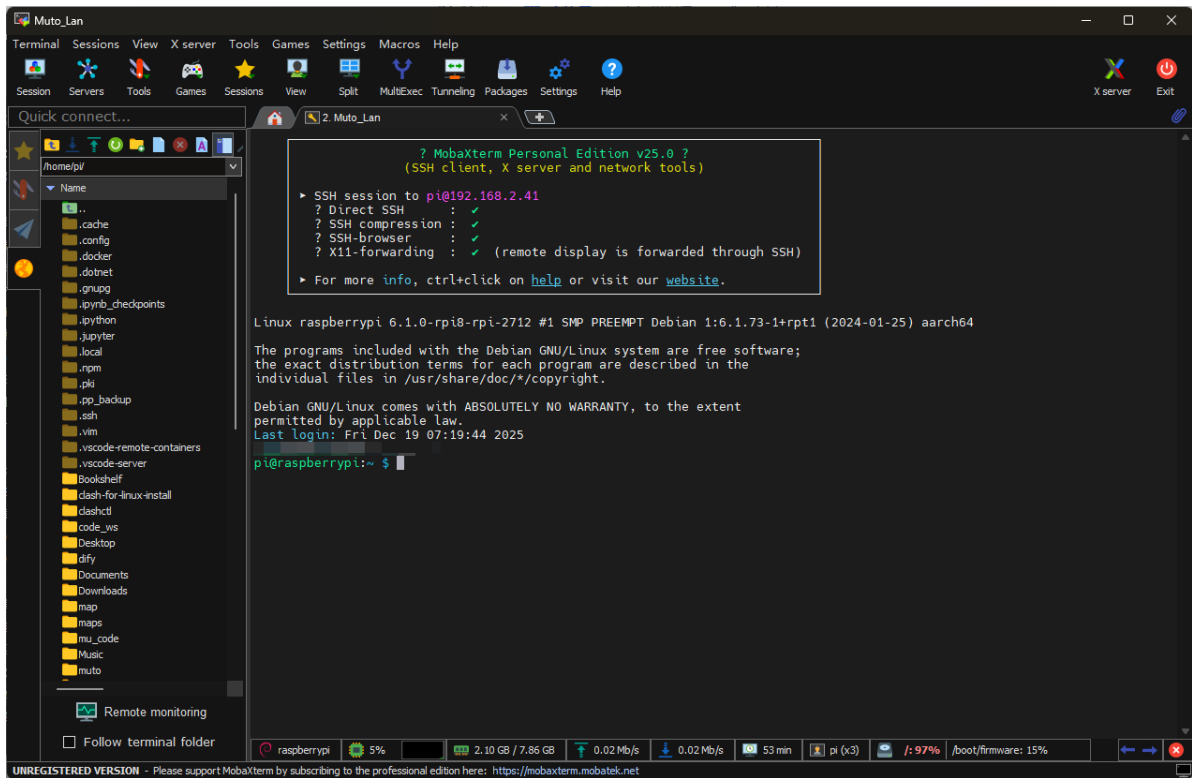
Red boxes and numbers highlight specific configurations:

- Red box 1: `export ALIBABA_API_KEY="sk-XXXXXXXXXXXXXXXXXXXX20"`
- Red box 2: `export VOICE_LANGUAGE="en"`

The Dify interface shows the configuration of the MutoLLM2.0 EN model. The interface includes a sidebar with options like "CREATE APP", "Workflow", "Chatflow", "Agent", and "Completion". The main area displays the configuration for the MutoLLM2.0 EN model, including the API key and the language setting.



This lesson requires MutoRS to be powered on. Following the steps in the previous chapters, start the large model terminal in Docker.



In this tutorial, we will use a plain text version of the large model, starting it with the `--no-voice` parameter. Using the `--no-voice` parameter eliminates the risk of being interrupted by external voice prompts during use!

```
cd muto-llm-2.0/ && ./rebuild_and_launch.sh --no-voice
```

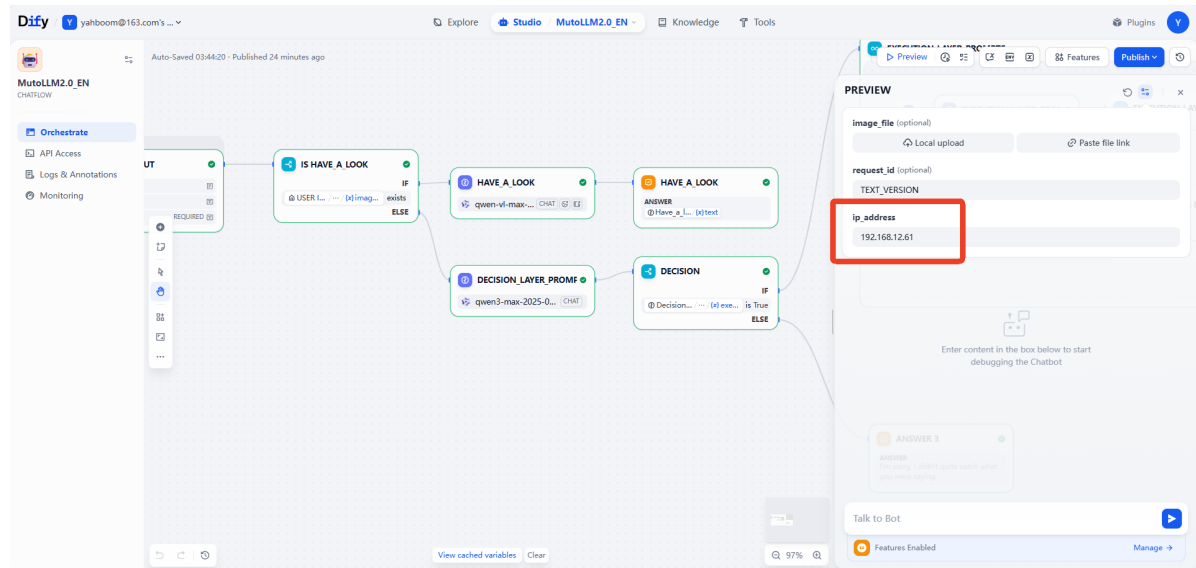
```

[muto_controller_node-1] 2025-12-19 09:48:12,313 - voice_module.voice_interface - INFO - Speech to text initialized successfully with alibaba provider
[muto_controller_node-1] 2025-12-19 09:48:12,313 - voice_module.core.text_to_speech - INFO - TTS Provider alibaba
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.core.audio_queue_manager - INFO - Audio queue processor started
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.core.audio_queue_manager - INFO - Audio queue processor started
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.voice_interface - INFO - Audio queue manager initialized successfully
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.voice_interface - INFO - All voice modules initialized successfully
[muto_controller_node-1] [INFO] [1766137692.315363643] [muto_controller_node]: Command executor initialized successfully
[muto_controller_node-1] [INFO] [1766137692.315920470] [muto_controller_node]: Attempting to prestart camera system...
[muto_controller_node-1] X Node not running: /camera/camera
[muto_controller_node-1] Current running nodes: ['/muto_controller_node', '/navigation_manager']
[muto_controller_node-1] [INFO] [1766137697.078459054] [muto_controller_node]: Camera service started successfully: Camera started successfully
[muto_controller_node-1] [INFO] [1766137697.085229984] [persistent_image_capture_3188dc8d]: Persistent image capture node initialized for topic: /camera/color/image_raw with Best Effort QoS
[muto_controller_node-1] [INFO] [1766137697.086642747] [muto_controller_node]: Persistent camera node initialized: Camera node initialized successfully
[muto_controller_node-1] [INFO] [1766137697.087635385] [muto_controller_node]: FastAPI application created successfully
[muto_controller_node-1] [INFO] [1766137697.149655757] [muto_controller_node]: API routes registered successfully
[muto_controller_node-1] 2025-12-19 09:48:17,149 - voice_module.core.voice_wake - INFO - Starting voice wake detector on port: /dev/myspeech
[muto_controller_node-1] 2025-12-19 09:48:17,154 - utils.serial_comm - INFO - Serial data receiving started
[muto_controller_node-1] 2025-12-19 09:48:17,154 - voice_module.core.voice_wake - INFO - Voice wake detector started successfully
[muto_controller_node-1] 2025-12-19 09:48:17,154 - voice_module.voice_interface - INFO - Wake detection started
[muto_controller_node-1] [INFO] [1766137697.155236161] [muto_controller_node]: Voice assistant auto-start: True
[muto_controller_node-1] [INFO] [1766137697.156658443] [muto_controller_node]: ROS2 publisher created successfully
[muto_controller_node-1] [INFO] [1766137697.157379009] [muto_controller_node]: Status timer and health check created successfully
[muto_controller_node-1] [INFO] [1766137697.157921559] [muto_controller_node]: Muto Controller Node initialized successfully
[muto_controller_node-1] [INFO] [1766137697.158425850] [muto_controller_node]: Starting FastAPI server on 0.0.0.0:8080
[muto_controller_node-1] [INFO] [1766137697.161239218] [muto_controller_node]: FastAPI server started successfully
[muto_controller_node-1] INFO: Started server process [221]
[muto_controller_node-1] INFO: Waiting for application startup.
[muto_controller_node-1] INFO: Application startup complete.
[muto_controller_node-1] INFO: Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)

```

Getting Started

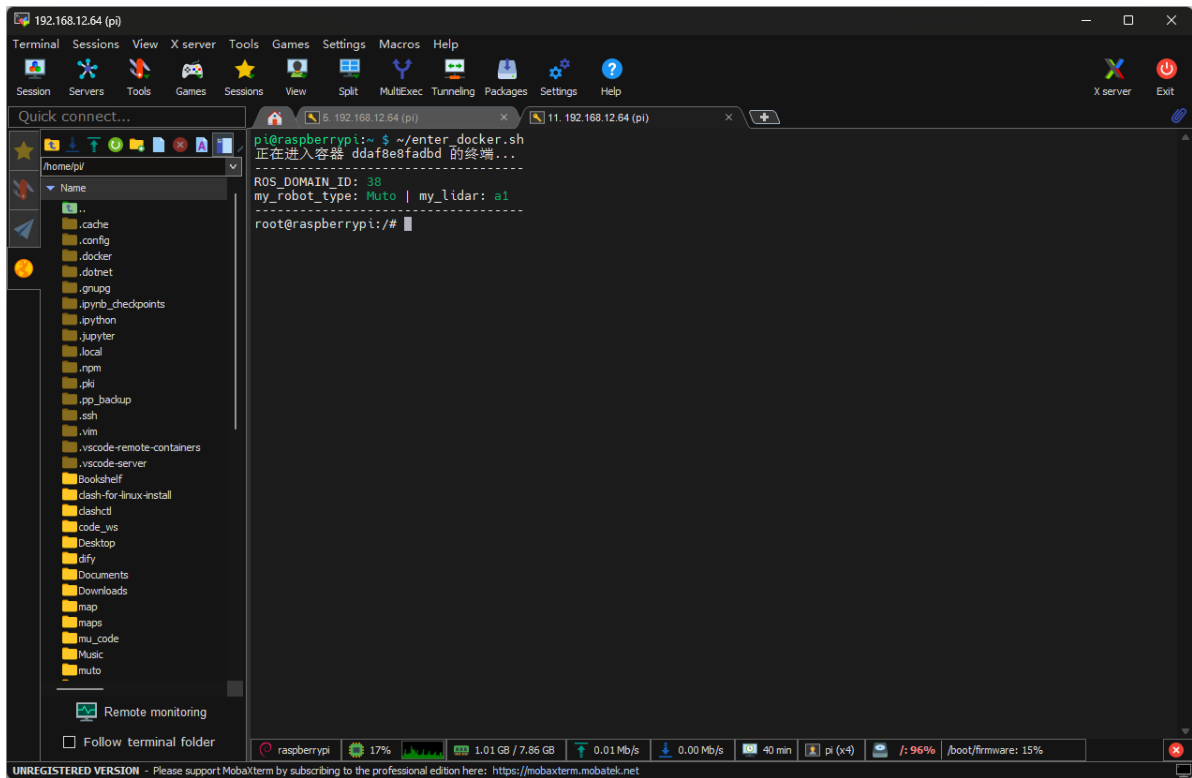
For the large text model, you need to enter Muto's IP address here, which is the address used to access the Dify page.



This section will explain the visual line-following function.

The visual line-following function requires opening a new terminal in the background, entering Docker, and running the visual line-following processing hub:

```
enter_muto_docker
```



In the new terminal, execute:

```
ros2 launch yahboomcar_linefollow follow_line_launch.py
```

As you can see, we've opened a window like this:

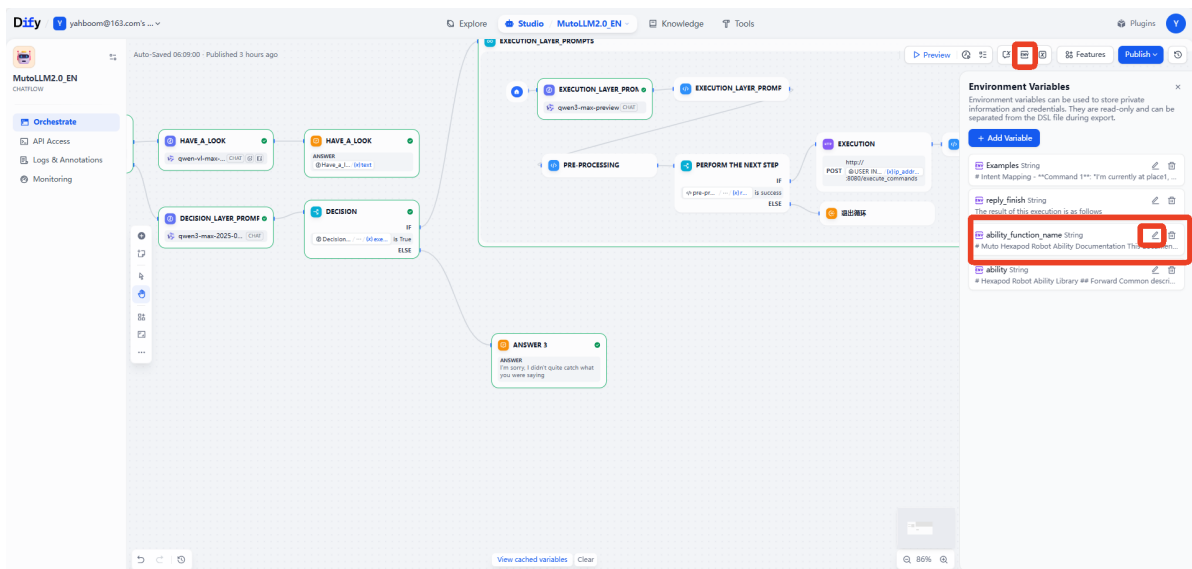


We use the left mouse button to select the yellow line and get its HSV range:

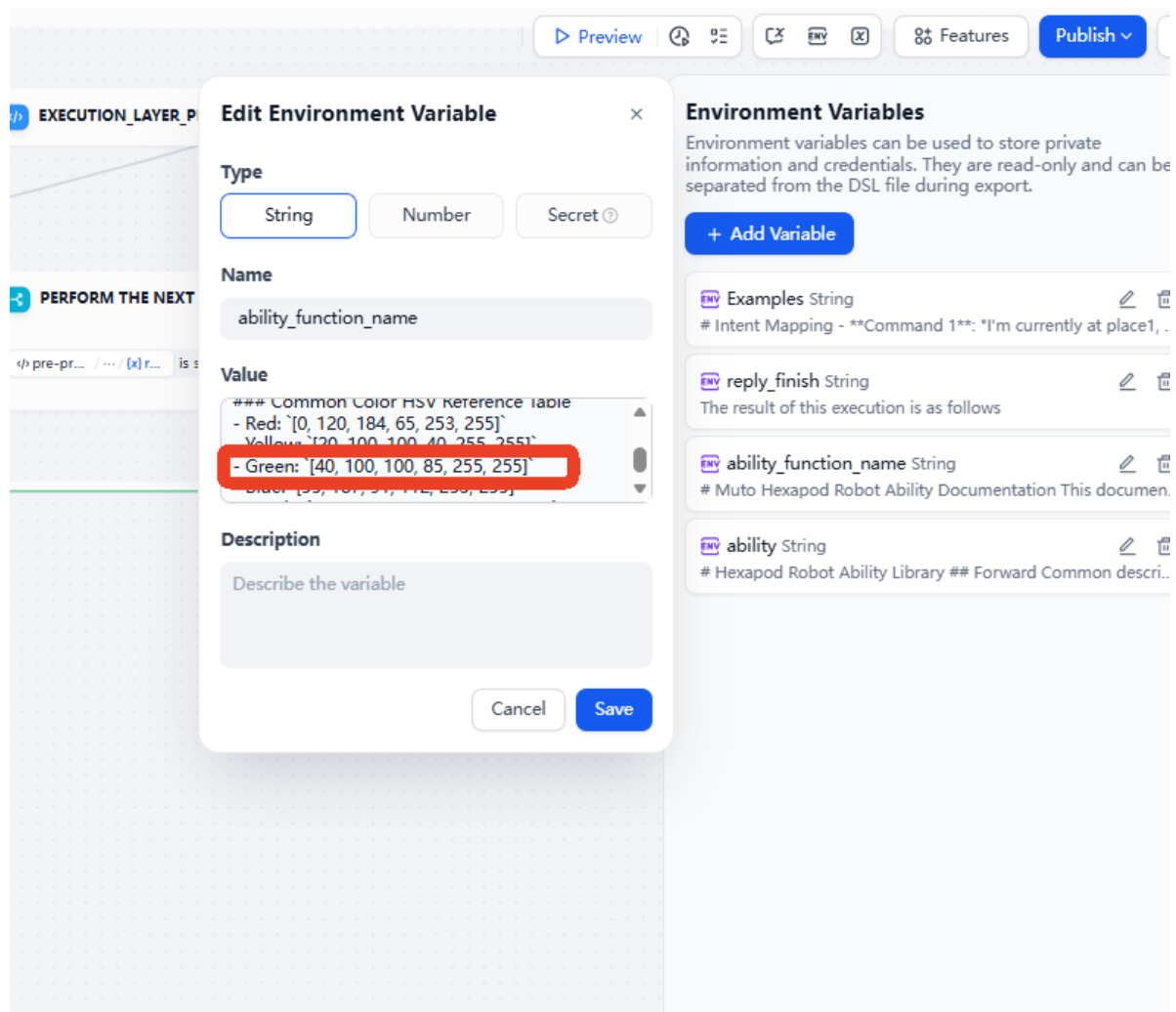


The HSV range in the top left corner is [35, 94, 96, 89, 253, 255]

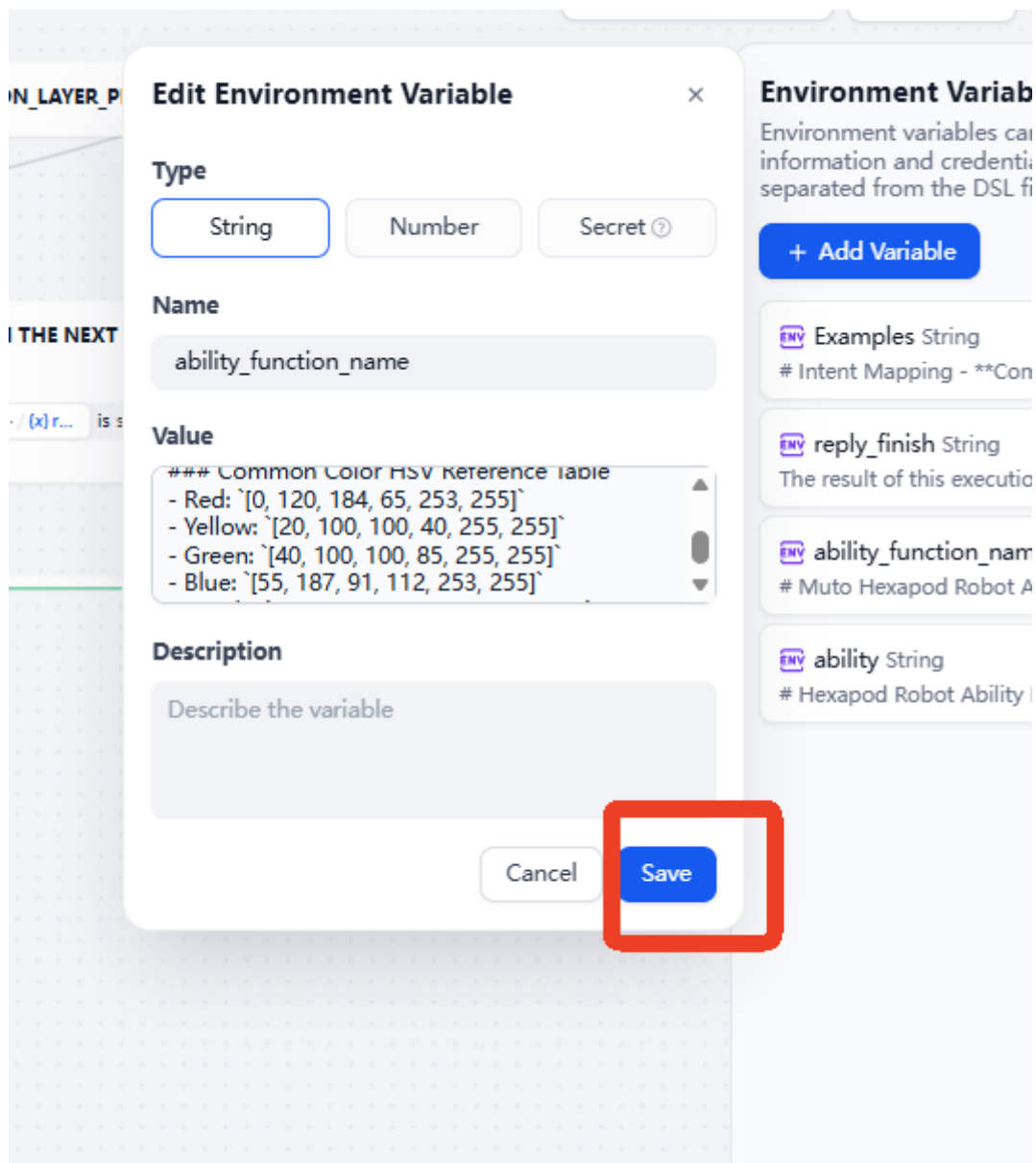
We copy this HSV range, open Dify, find the env icon in the upper right corner, and edit our ability_function_name variable.



Edit the HSV value here, changing the yellow HSV value to the one we just obtained.



After making the changes, click save.



The command we use in this section is:

"Start autopilot along the line of color xx in front of you."

As you can see, Muto will announce the voice prompt and then begin to move automatically along the line.

